

Rui Rodrigues – Curriculum Vitae

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ABOUT ME

I have a keen interest in fields like operations research, optimization, simulation and machine learning. This has led me to try different branches of work in different contexts but always in related areas.

EDUCATION

09/21 – today **Faculdade de Engenharia da Universidade do Porto (FEUP)** **Portugal**
Bachelor's degree in industrial engineering and management – finished
Master's degree in industrial engineering and management - in progress

EXPERIENCE

10/24 – today **INESC TEC, Research Fellowship, Master's Thesis** **Portugal**

- Development of algorithms and optimization methods for benefit and energy allocation in renewable energy communities, as well as optimal portfolio of members and loyalty mechanisms.
- Energy management tool for ports, with the development of models for forecast and optimization of different assets, including a MILP that allows the use of reefers and thermal energy storage.
- Work documented in ~5 journal papers and ~3 conference papers (published, under revision or submitted)

02/26 – 04/26 **LTPLabs, Business Analyst Trainee** **Portugal**

03/24 – 09/24 **ACM FEUP, Development Manager** **Portugal**

06/24 – 07/24 **INESC TEC, Summer Internship** **Portugal**

- **Data-driven approach** for electricity forecast in highly electrified seaports, incorporating operations data, with different **machine learning** techniques, handling the complete implementation pipeline.
- Awarded **Best Summer Internship at CPES** (Centre for Power and Energy Systems).

12/23 – 06/24 **INEGI, Curricular Internship** **Portugal**

- Developed a Vehicle Routing tool to generate routes for a retail company. As a result of positive feedback, I was part of a new project at INEGI regarding ideal fleet **size and mix**.

10/21 – 01/24 **Tutor at Centro Social Paroquial de S. Nicolau.** **Portugal**

SKILLS

- **Python**, Apps, OOP, Pyomo, Matplotlib, Numpy, PyTorch, Pandas, and a lot more.
- **Relational Databases**, including SQL and Bash.
- **ML/AI**, Supervised, Unsupervised and Reinforcement Learning, from traditional to SOTA.
- **Optimization**, Linear, Non-linear, Stochastic/Probabilistic, Bilevel/Hierarchical, Multiobjective
- **Simulation**, Discrete Event and Time, Agent-based. AnyLogic and SimPy

PROJECTS

Integrated Car Rental Fleet Management: A Matheuristic Approach **20/20**

- Developed a matheuristic approach for pricing and deployment on car rentals, with a strategy scalable to large and realistic size networks (generated to be much larger than benchmarks).
- This project was the best of the course.

Dynamic Promotion and Restocking Optimization for Multi-Product Perishable Inventory **20/20**

- Developed an event-driven environment and employed a QR-DQN Reinforcement Learning approach able to beat heuristics, while showing value from combining pricing and restocking.
- This project was the best of the course.

Neural Network Forecast for Sparse Fashion Retail Sales (Beating Company Baseline) **20/20**

- This project was the best of the course.

Semantic Segmentation of Maritime Thermal Imagery Using Lightweight Attention-Based U-Nets **20/20**

- This project was the best of the course.

Sonae MC Network Optimization

- Sonae MC challenge to design the transportation network, as well as fleet size and mix to minimize total costs and maximize efficiency responding to a growing number of proximity stores.
- This project was the winner of the challenge.

HONORS AND AWARDS

2021 - today Several case studies and challenges such as: **finalist team at Tecstorm 2026; winner team at Rangel Case Study 2025; best speaker at APA's model parliament 2024; last phase of PwC Consulting Challenge 2023; 2nd place team at TIMES LQ 23; 3rd place team at the 1st edition of Portugal Case Study**